SENG 696

A Diplomacy-Style Negotiation Game with Autonomous Agents

Design Document

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1 Business Pitch / Purpose

A diplomacy, turn-based strategy game in which a set of autonomous software agents and a human player each control a regional power competing for territory of a digital board game map. There are no dice or hidden randomness but rather communication with other powers to resolve a deterministic battle between units. The game will be based off a loose version of the rules for the popular game Diplomacy but with some slight simplifications to make agent-based interaction easier. The purpose is to provide a playable negotiation game in which a human competes against autonomous, self-interested nations. Games are one of the leading forms of modern media and the adaptation of classic board games to digital media is rare. By leveraging agents, we can make a game world that allows for self-interested parties that can negotiate and maintain private relations.

The player will interact with the world in the same way the Agent powers will by negotiating, forming and breaking alliances, and ordering units. The world will respond by the decisions of other powers that each are self-serving, and outcomes dictated by an authoritative game master agent that broadcasts the game state at the end of the turn.

Figure 1 shows the game architecture and how the human will interact with other players and the game world.

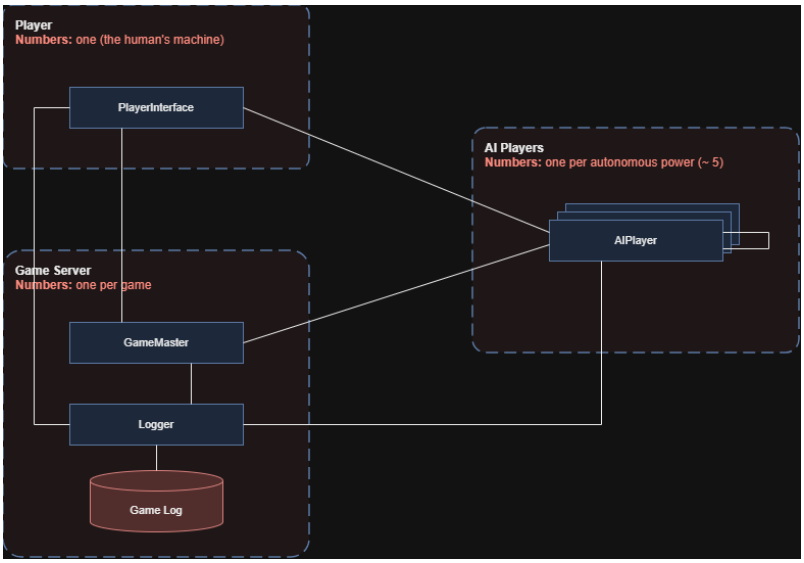


Figure 1: Game architecture

2 Game Overview

The game is a simplified version of Diplomacy where territory is the main resource with support being the only thing that can be traded. The game board will consist of a map that has 45 territories with some supply centres that will generate troops. There will be approximately 5 different powers, each with 2–3 troops and supporting supply centres. One of the powers will be controlled by the human client agent and the rest will be controlled by "AI" agents.

2.1 Units and Ownership

A territory can hold at most 1 unit. A nation owns a territory by being the last to occupy it (painting that territory your colour). Every 3 turns, each nation will grow or shrink their armies to the number of supply centres it owns. The spawn location of these units will be the supply centre that has recently been added.

2.2 Orders

Each turn every unit receives exactly one order: hold, move to adjacent territory or support an adjacent unit's move (+1 strength).

2.3 Turn Cycle

All powers act simultaneously and no one sees the move another power plans to perform that turn until the game master agent broadcasts the turn result. All phase transitions (negotiation → order and order → negotiation) will be player guided.

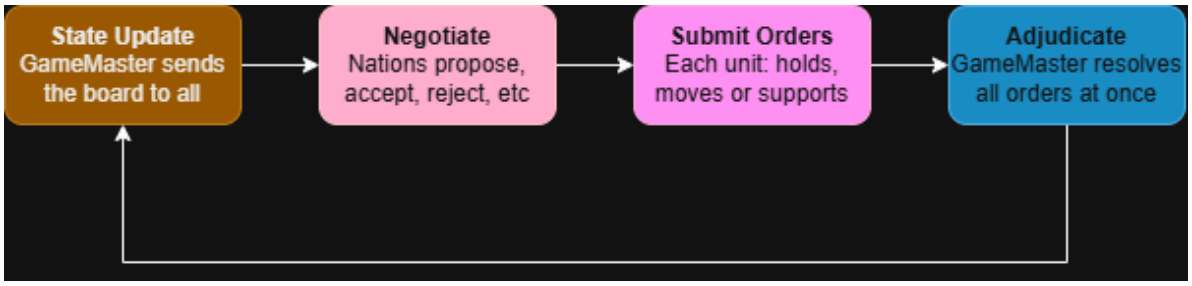


Figure 2: Turn cycle

2.4 Adjudicate

Deterministic resolve of all movements and orders. A move in which a power attempts to move into a place where a foreign power's unit is placed will result in a combat. The winner of this combat is the one with the higher overall power after adding support from allied power troops. Ties will result in units bouncing off each other and losing will take that unit off the board.

2.5 Negotiation

Nations exchange private structured messages using a fixed set of possible communications: propose alliance, propose non-aggression, request support, accept, reject and inform. These are the only decisions player and non-player powers can engage in with other agents when it comes to negotiating.

2.6 Victory

The first nation to own a majority of the total of supply centres wins or if the turn limit is reached, then the one with the most supply centres.

3 Assumptions

- The Logger stores accepted agreements and the board and results of every turn, nothing else.
- No timers. The negotiation phase ends when the human clicks End Negotiations; the order phase ends when every power has submitted orders.
- AI powers negotiate during the phase and submit their orders as soon as the order phase starts, so the human sets the pace of the game.
- At all stages of the game the players can see the board.
- Units must play valid moves only.
- With no supply centres and units, the player is eliminated.
- All AI personalities are set at the beginning of a game and remain.
- Each game is unique and doesn't affect others.

4 Requirements

- The player will control 1 nation through a GUI showing the board, unit positions, ownership and current phase.
- The player shall be able to communicate in the same way as the non-player powers.
- The player will be capable of giving each unit their orders.
- The player will receive a full breakdown of all alliances formed and promises broken relating to them.
- The system will have a single authoritative board state.
- The system shall run the turn cycle with messages sent out of order treated as void.
- The system will resolve all orders simultaneously according to the rules.
- The system will handle the spawning and removal of units.
- The system shall declare a game winner.
- Each power shall receive the board state and adjudication results each turn.
- Each power shall select objectives and generate legal orders from their own beliefs.
- Each power shall initiate proposals to each nation and evaluate incoming proposals, replying with accept or reject.
- Each power shall record commitments made and received, compare with the adjudicated outcomes and maintain a trust score for every nation.
- Each power shall decide each turn whether to break or follow through with its promises by weighing the cost and gain.
- Each power will be configurable with a personality profile that changes how it values alliances, risk and promise-keeping.
- The system shall log every negotiation message and adjudication result.
- The system shall export a game recap that includes winner, board state per turn and a relationship log.
- No power may read another power's state or messages or relationships.
- The player will be constrained to the same rules as the non-player powers.
- The order phase and negotiation phase will be limited in time by the human player's discretion.

5 Wishlist

- AI players get to negotiate through LLM interface instead of strict type of negotiations and orders based off game logic.
- More types of units and interactions in the game.
- Timer-based combat and negotiations.
- Complex resource management and map structures.